Academic & Clinical Graduation Project | 2026

CareElder Platform

Department of Software Engineering & Information Technology

ENGINEERING SPECIFICATION & USE CASES

CareElder (رفيق العافية)

Intelligent Healthcare, Medication Adherence & Clinical Monitoring Platform for Seniors

> A rigorous architectural and functional specification detailing verified Use Cases, clinical problem formulations, accessible geriatric user experience design, and cross-stakeholder workflows linking senior patients, family caregivers, and attending physicians.

System Architecture:	Offline-First Architecture & Delta Sync	**Technology Stack:**	Flutter + FastAPI + Computer Vision OCR
Target Demographics:	Chronic Disease Patients & Elderly Seniors	**Accessibility:**	Geriatric Accessibility (WCAG AAA)
Document Type:	Engineering Specification & Use Cases	**Release / Status:**	Version 2.6 (Streamlined Minimalist Edition)

Engineering Team: CareElder Graduation Project Team - Class of 2026

Date: 2026

00. Executive Summary & Table of Contents

Strategic platform vision, primary KPIs, and comprehensive 11-page document directory

Executive Summary

CareElder (رفيق العافية) is a specialized digital health platform designed for aging seniors and chronic patients. Departing from complex commercial apps, CareElder implements an **Offline-First Architecture** with simplified interfaces tailored to geriatric motor and cognitive realities. The system synchronizes the senior patient, family caregiver, and physician via hardware-level exact alarms, multimodal prescription OCR, zero-typing colloquial voice logging, pill inventory management, and longitudinal clinical analytics.

> 96%	**< 3.5s**	**100%**	**Zero**
Target Adherence Rate	AI Prescription OCR Speed	Offline Alarm Guarantee	Typing (100% Voice UX)

Document Directory (11 Dedicated Engineering Pages)

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01. Root Problem Statement & Clinical Challenges

Global health metrics, polypharmacy hazards, and the silent crisis of geriatric non-adherence

The Clinical & Economic Crisis (WHO Benchmark Data)

According to the **World Health Organization (WHO)**, over **50%** of patients with chronic diseases (hypertension, diabetes, cardiovascular failure) fail to take prescribed medications correctly. This widespread non-adherence causes **33% to 45% of all emergency hospital readmissions** and significant preventable complications such as stroke and renal failure, imposing severe emotional and financial burdens on families and healthcare systems.

1. The Senior Patient Dilemma

- **Extreme Polypharmacy:** Managing 5 to 9 daily pills with differing intervals exceeds normal cognitive working memory in aging seniors.
- **Sensory & Motor Impairment:** Presbyopia prevents reading tiny labels; tremors and arthritis make phone keyboards unusable.
- **Temporal Confusion:** Constant disorientation regarding whether today's dose was taken or merely remembered from yesterday.
- **Commercial App Overload:** Existing applications are cluttered with ads, small fonts, and convoluted multi-step workflows.

2. Caregiver Burnout & Distance Gap

- **Work-Life Strain:** Adult working children suffer constant anxiety during office hours, repeatedly calling without getting clear confirmation.
- **The "Courtesy Bias" Trap:** When asked "Did you take your pills?", seniors frequently say "Yes" to avoid feeling burdensome.
- **Chaotic Inventory Management:** Discovering that insulin pens or heart medication strips ran out at 11:00 PM on a weekend.
- **Clinical Recall Void:** Doctors must guess therapy efficacy based on inaccurate patient memory during consultations.

Comparative Analysis: Status Quo vs. CareElder Solution

Evaluation Metric	Commercial Apps & Traditional Habits	CareElder Engineering Innovation
Medication Alarms	Subtle banner notifications that dismiss instantly and fail offline.	**Full-Screen Intent** override with loud audio & offline hardware alarms.
Prescription Entry	Tedious manual text typing in English; high user friction.	**Multimodal AI OCR** converts paper prescriptions to schedules in one click.
Vital Signs Logging	Manual numeric keypad entry with frequent transpositions.	**100% Voice-driven colloquial NLP** with automatic clinical color triage.
Inventory Governance	Manual guesswork and frequent emergency stock-outs.	**Automatic dose decrementing** with proactive 5-day WhatsApp refill generation.

02. Platform Value Pillars & What The System Offers

Five core engineering pillars delivering holistic, preventative senior healthcare

Architectural Philosophy: Empathetic Geriatric Engineering

CareElder is engineered as a **continuous preventative healthcare ecosystem** bridging patients, families, and clinicians. The platform is grounded in five core architectural pillars that ensure offline reliability, ease of use, and verified clinical utility.

1. Exact Offline Alarms

Leverages OS-native AlarmManager and Full-Screen Intent to trigger critical wakeups even when the phone is locked, silenced, or disconnected from the internet.

2. AI Prescription OCR

Computer Vision model tailored for doctor handwriting extracts drug names, strengths, frequencies, and food correlations with sub-4s latency.

3. Zero-Typing Voice UX

Colloquial speech-to-text pipeline extracts numeric vital signs (BP, Glucose) and evaluates clinical ranges instantly without keyboard interaction.

4. Caregiver Cloud Escort

Real-time timeline displaying dose verification timestamps and an automated 3-tier escalation ladder for unconfirmed medications.

5. Clinical Reports & Inventory

Automated pill decrementing preventing therapy lapses, and longitudinal PDF reports with encrypted QR access for attending physicians.

CareElder Closed-Loop Daily Care Execution Pipeline

1. Scan Prescription	2. Local Schedule	3. Wakeup & Voice	4. Sync & Inventory	5. Clinical Audit
AI Vision extracts drug entities from paper Rx	Exact hardware alarms set inside local database	Full-Screen Intent wakes phone with spoken guidance	Live caregiver update & pill inventory decrement	Longitudinal PDF trends & encrypted QR code review

03. Stakeholder Ecosystem & User Personas

Detailed architectural profiling of target personas and cross-stakeholder interactions

Haj Ibrahim (Age 73)

Chronic Patient

Clinical Condition: Type-2 Diabetes & Hypertension; takes 6 daily medications.
Digital Literacy: Low; struggles with small fonts and clumsy typing.
Core Needs: High-contrast giant buttons, audible voice prompts in warm English/Arabic, and speaking vital measurements without touching keypad digits.

Active Interface: Senior Mode (High Contrast, 24sp fonts, Zero Typing).

Mona (Age 39)

Family Caregiver

Social Profile: Full-time working mother living 35 km away from her father.
Core Pain Point: Chronic work-hours anxiety and fear that her father forgets his heart pills.
Core Needs: Live dashboard displaying intake confirmations, automatic escalation alerts if doses are neglected, and remote pill refill automation.

Active Interface: Caregiver Hub (Live Timeline, Rx Scanner, Refill Alerts).

Dr. Tarek (Age 52)

Attending Doctor

Clinical Context: Evaluates senior patients quarterly to calibrate medication dosages.
Core Pain Point: Patient recall bias ("I think my pressure was around 130...").
Core Needs: Objective 90-day longitudinal adherence scores, BP/Glucose trend curves, and missed dose logs delivered via 1-tap QR scan or clinical PDF.

Active Interface: Clinical Web Portal & Instant QR Report Viewer.

Cross-Stakeholder Interaction & Data Flow Matrix

Actor	Primary Input to System	Delivered Output & Clinical Value	Channel
Senior Patient	Dose confirmation (1-tap), voice-spoken vitals	Audible guidance, giant visual cues, independence without typing	Senior Mode UI
Family Caregiver	Prescription photo, schedule adjustments, inventory setup	Real-time compliance feed, escalation alerts, refill alerts via WhatsApp	Caregiver Hub
Attending Doctor	Prescription updates, biomarker safety boundaries	Longitudinal PDF trends, adherence rate %, missed dose forensic log	QR Scan / PDF Export

04. Use Case 01: Smart Medication Alarm & Acknowledgment

Offline-First Exact Hardware Alarms, Full-Screen Intent Wakeup & Zero-Latency Acknowledgment

Use Case ID: `UC-01`	**Primary Actor:** Senior Patient	**Priority:** Critical (Must Have)	**Execution:** 100% Offline-Ready

Operational Context & Goal

Guarantee that the senior patient is reliably awakened and alerted at the exact scheduled second, bypassing system silence and screen locks, with audible spoken instructions, high-contrast imagery of the specific pill, and instant offline acknowledgment.

Main Success Scenario (Step-by-Step Execution Flow)

1. **Exact Alarm Hardware Trigger:** The OS-level Exact Alarm Scheduler (Android AlarmManager / iOS Background Task) triggers at the precise scheduled timestamp, independent of battery optimizations or internet connectivity.
2. **Full-Screen Intent Wakeup:** The display wakes up completely over the locked screen with high-contrast styling (AAA), pulsating visual cues, and an escalating, pleasant acoustic chime.
3. **Audible Voice Guidance (TTS):** The system speaks aloud: "*Time for Concor 5mg heart tablet after lunch, Haj Ibrahim*" while displaying an enlarged high-resolution image of the pill blister.
4. **One-Tap Acknowledgment:** The patient taps the giant button **[Pill Taken Now]**. The chime stops immediately, and the app speaks a confirmation message.
5. **Local Persistence & Delta Sync:** The event is committed locally to the Isar/SQLite database with a microsecond timestamp, and a compact encrypted delta payload is dispatched to the cloud for caregiver synchronization.

Alternative Flow 1: 10-Min Snooze

If the patient is eating or indisposed, they tap [Snooze 10 Mins]. The system reschedules the alarm for exactly 10 minutes later. Up to 3 snoozes are permitted before caregiver escalation initiates.

Alternative Flow 2: Ignored / Missed Alarm

If the alarm chimes for 60 seconds without touch, it transitions to a pending state. If unconfirmed after 30 minutes, the Cloud Adherence Watcher triggers a Tier-1 alert to the caregiver's phone.

05. Use Case 02: AI Prescription OCR & Medical NER

Multimodal Vision-Language Extraction of Handwritten Prescriptions into Structured Regimens

Use Case ID: `UC-02`	**Primary Actor:** Caregiver / Senior	**Priority:** Critical (Must Have)	**Target Latency:** < 3.5 Seconds

Operational Context & Goal

Enable caregivers to snap a photo of paper prescriptions (whether typed or handwritten by doctors) and have an AI vision engine extract drug entities, strengths, frequencies, and food correlations into an automated alarm schedule with human verification for clinical safety.

Computer Vision & Natural Language Processing Pipeline

1. **Guided Optical Capture:** The camera UI renders a dynamic rectangular framing guide with on-device gyro-assisted angle detection to prevent perspective skewing and motion blur.
2. **Adaptive Image Pre-Processing:** Automatic document contour detection, perspective unwarping, shadow suppression, and adaptive thresholding optimize handwriting legibility.
3. **OCR & Medical Entity Recognition (NER):** The processed image passes to a multimodal vision pipeline with medical dictionary fuzzy matching to parse: `[Drug Name, Strength_mg, Frequency, Food_Relation, Duration]`.
4. **Interactive Verification Card (Human-in-the-Loop):** The extracted medications populate an editable confirmation card. Fields with confidence scores below 85% are highlighted for instant tap-confirmation.
5. **Automated Schedule & Inventory Synthesis:** Upon approval, alarms are programmatically synchronized with the senior's phone, and initial pill inventory quotas are registered automatically.

Structured Output Schema (JSON Contract)

```
{
  "drug_trade_name": "Glucophage",
  "dosage_mg": 500,
  "frequency_per_day": 2,
  "meal_relation": "after_meals",
  "times": ["08:30", "20:30"],
  "inventory_pills": 60,
  "confidence_score": 0.96
}
```

06. Use Case 03: Voice Vitals Logging & Clinical Triage

Zero-Typing Speech-to-Text Vitals Extraction with Instant Clinical Boundary Classification

Use Case ID: UC-03	**Primary Actor:** Senior Patient	**Priority:** Critical (Must Have)	**Languages:** Colloquial Arabic & English

Operational Context & Goal

Empower the elderly patient to record daily biometric readings (blood pressure, fasting/random glucose, pulse) by speaking numbers naturally without typing a single digit. The engine parses the values, classifies them according to clinical guidelines, and notifies caregivers if critical.

Execution Sequence & Clinical Evaluation Protocol

1. **Single-Tap Voice Initiation:** The senior taps the large microphone button in the center of the dashboard. An audio chime indicates that the system is actively listening.
2. **Natural Spoken Utterance:** The patient speaks casually: "*My pressure today is 135 over 85*" or "*Fasting sugar is 120*".
3. **Speech-to-Text & Numeric Parsing:** The on-device/cloud STT engine transcribes speech, extracts metric types and numeric values, and speaks an audio confirmation back to the patient.
4. **Instant Clinical Decision Triage:** The system matches values against standard clinical guidelines (AHA/WHO) and displays clear status feedback immediately:

1. Optimal Band	2. Borderline Band	3. Critical Band
BP < 120/80 Fasting Glucose 70-110	BP 130-139 / 80-89 Fasting Glucose 111-140	BP ≥ 180 or ≤ 90 Glucose < 70 or > 250
"Normal stable reading"	**"Mildly elevated, monitor salt"**	**"Instant Caregiver Alert!"**

Critical Band Intervention Protocol

If a severe anomaly is detected (e.g. acute hypoglycemia < 60 mg/dL or hypertensive urgency > 180 mmHg), the app speaks immediate soothing guidance: "*Haj Ibrahim, please sit down calmly and drink some fruit juice*" while simultaneously pushing an alert to the caregiver's device.

07. Use Case 04: Remote Caregiver Supervision & Real-Time Sync

Real-Time Cloud Synchronization, Multi-Tier Adherence Escalation & Remote Schedule Governance

Use Case ID: `UC-04`	**Primary Actor:** Family Caregiver	**Priority:** Critical (Must Have)	**Sync Protocol:** WebSocket & FCM Delta Sync

Operational Context & Family Peace of Mind

Provide working or distant family members with 24/7 visibility into the senior's medication status and vital signs via a reactive mobile dashboard, backed by an automated escalation engine that intervenes whenever doses are missed without burdening the senior.

Caregiver Mobile Hub Components

- **Live Medication Timeline:** Dynamic feed displaying today's regimen: Taken at 08:14 AM, Snoozed, or Upcoming.
- **Vitals Stream:** Instant cards displaying latest blood pressure and blood sugar readings alongside 7-day rolling averages.
- **Personalized Audio Notes:** Caregivers can record warm voice memos from grandchildren that play with the alarm to motivate the senior.
- **Live Intake Confirmation:** Real-time push updates delivering total peace of mind during office hours.

Automated 3-Tier Adherence Escalation Ladder

> **Minute 0:** Exact hardware alarm sounds on senior's phone with audible spoken pill guidance.

> **Minute +15:** Gentle secondary chime on patient device if unconfirmed.

> **Minute +30:** Tier-1 Care Alert pushed to caregiver: *"Notice: Your father has not confirmed his morning heart dose."*

> **Minute +60:** Tier-2 Urgent Alert with an embedded 1-tap call button encouraging immediate contact.

Remote Prescription & Schedule Governance

Caregivers can adjust dosages or add newly prescribed medications directly from their own smartphones. All schema modifications sync instantly to the senior's phone without requiring any complex technical interaction from the senior.

08. Use Case 05: Automated Pill Inventory & Low-Stock Alerts

Predictive Medication Stock Burn-Rate Calculation & One-Tap Pharmacy Reordering

Use Case ID: `UC-05`	**Primary Actor:** Caregiver / Inventory Engine	**Priority:** Important (Should Have)	**Objective:** Eliminate chronic medicine shortages

Operational Context & Goal

Completely eradicate sudden medication stock-outs by tracking remaining tablet counts in real-time, forecasting exact depletion dates based on consumption patterns, and triggering proactive caregiver warnings 5 days prior with a pre-formatted pharmacy refill order.

Inventory Tracking & Predictive Replenishment Workflow

1. **Batch Registration:** When registering a new medication (via OCR scan or manual entry), the caregiver inputs the total pill count (e.g. 2 blister strips = 20 tablets of Concor 5mg).
2. **Synchronized Auto-Decrement:** Each time the senior confirms a dose, the system automatically decrements the inventory quota across both local and cloud databases.
3. **Depletion Date Forecasting:** The algorithm calculates the precise depletion milestone: `Depletion_Date = Today + (Pills_Remaining / Daily_Doses)`.
4. **Proactive 5-Day Threshold Warning:** When remaining stock falls below 5 days of therapy, an alert notifies the caregiver: "*Stock Alert: 5 days of heart medication remaining for Haj Ibrahim.*"
5. **1-Tap Neighborhood Pharmacy Reorder:** The caregiver taps [Generate Order], which compiles a structured refill manifest ready to send via WhatsApp to the family's preferred local pharmacy with a single tap.

Sample Automated Pharmacy WhatsApp Manifest

```
Hello Care Pharmacy, please prepare refills
for Patient Ibrahim:
1. Concor 5mg (Box of 30 tabs)
2. Glucophage 500mg (Box of 60 tabs)
Delivery Address: 12 Tahrir St, Apt 4.
```

Medication Expiration Date Guard

Caregivers can capture the expiration date stamped on the blister strip. The system monitors shelf life and flags medications 30 days before expiration, preventing the inadvertent intake of expired pharmaceuticals.

09. Use Case 06: Longitudinal Clinical Reporting for Doctors

Longitudinal Adherence Trend Analytics, Clinical Vital Graphing & Encrypted QR Doctor Portals

Use Case ID: UC-06	**Primary Actor:** Doctor / Caregiver / Patient	**Priority:** Critical (Must Have)	**Formats:** Clinical PDF & Encrypted QR

Clinical Value Proposition

Transform clinical consultations from subjective, memory-based guesswork into precision data reviews. CareElder compiles 30 to 90 days of verified adherence data and vital sign readings into an executive clinical report that empowers physicians to make informed therapeutic adjustments.

Clinical Longitudinal Report Architecture

1. Cumulative Adherence Score %

A verified metric computing on-time medication intake versus prescribed schedules (e.g. 94.5% adherence over the past quarter).

2. Biometric Longitudinal Curves

Continuous trend graphs mapping systolic/diastolic blood pressure and fasting/postprandial glucose over 30 or 90 days.

3. Missed Dose Forensic Log

Granular incident logs documenting unconfirmed doses, timestamps, and recorded reasons (forgetfulness, stock shortage, or repeated snoozing).

4. Active Multi-Drug Regimen

Comprehensive listing of brand names, active scientific entities, and dosages to prevent dangerous prescribing duplications.

Zero-Install Clinical QR Consultation Scan

Instead of carrying cumbersome paper binders, the caregiver presents a temporary **encrypted QR code**. The physician scans it with their clinic smartphone or workstation camera, immediately opening a secure, interactive web dashboard without installing any software.

Clinical Efficacy & Impact on Dosage Decisions

Clinical research demonstrates that objective adherence reporting reduces diagnostic consultation time by 40% and prevents clinicians from inadvertently increasing drug dosages when poor control is actually caused by missed pills rather than pharmacological tolerance.